

Optimization applied 1:

Maximum likelihood estimation

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November 25, 2025

Maximum Likelihood Estimation: introduction

MLE: Intuition (Bernoulli)

Let $Y \sim \text{Bern}(p)$.

- Suppose we observe 6 i.i.d coin flips (1 for heads, 0 for tails):

$$\mathbf{y} = \{1, 1, 1, 0, 1, 0\}$$

- Denote each flip as $y_1, y_2, \dots, y_6$, where $y_1 = 1$ and $y_6 = 1$.

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Recall: the probability mass function (PMF) of a Bernoulli r.v. is:

$$f(y; p) = p^y(1 - p)^{1-y}, \quad y \in \{0, 1\}$$

The probability of 1 is p , while the probability of 0 is $(1 - p)$.

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Overarching Goal: Estimate p given a vector of observed data.

What parameter $\hat{p}$ maximizes the probability of observing our particular sample?

MLE: Intuition (Bernoulli)

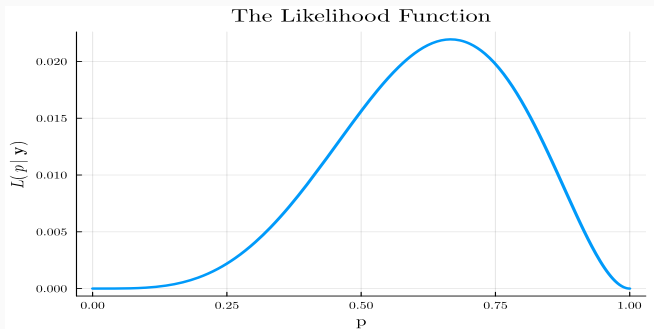
- Assume y_i are independent
- The likelihood of observing $\mathbf{y} = \{1, 1, 1, 0, 1, 0\}$ is:

$$L(p \mid \mathbf{y}) = \prod_{i=1}^6 p^{y_i} (1 - p)^{1-y_i}$$

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MLE: Log likelihood function (Bernoulli)

$$L(p \mid \mathbf{y}) = \prod_{i=1}^n p^{y_i} \cdot (1 - p)^{1-y_i}$$

Necessary condition for maxima:

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$$\begin{aligned} L(p \mid \mathbf{y}) &= \prod_{i=1}^n p^{y_i} \cdot (1-p)^{1-y_i} \\ \log(L(p \mid \mathbf{y})) &= \sum_{i=1}^n \log(p^{y_i}) + \sum_{i=1}^n \log(1-p^{1-y_i}) \\ &= \sum_{i=1}^n y_i \log(p) + \sum_{i=1}^n (1-y_i) \log(1-p) \end{aligned}$$

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Necessary condition for maxima:

$$\frac{\partial \log(L(p \mid \mathbf{y}))}{\partial p} = 0$$

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Necessary condition for maxima:

$$\begin{aligned}\frac{\partial \log(L(p | \mathbf{y}))}{\partial p} &= 0 \\ \frac{\sum_{i=1}^n y_i}{p} - \frac{\sum_{i=1}^n (1 - y_i)}{1 - p} &= 0\end{aligned}$$

MLE: Log likelihood function (Bernoulli) maximized

Note: In this simple example we can solve for p !

$$\frac{\partial \log(L(p | \mathbf{y}))}{\partial p} = 0$$

$$\frac{\sum_{i=1}^n y_i}{p} = \frac{\sum_{i=1}^n (1 - y_i)}{1 - p}$$

Using: $y = \{1, 1, 1, 0, 1, 0\}$

$$\frac{4}{p} = \frac{2}{1 - p}$$

$$4 - 4p = 2p$$

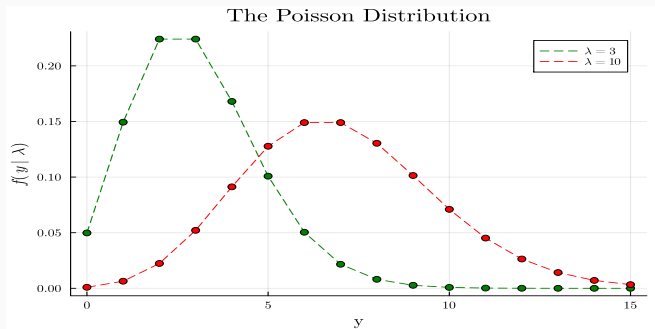
$$-6p = -4$$

$$\hat{p} = \frac{4}{6}$$

Let's see how to implement MLE in Julia!

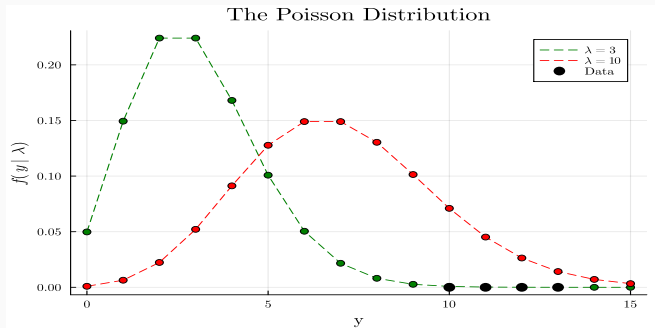
A task for you: the Poisson Distribution

$$f(y | \lambda) = \frac{e^{-\lambda} \lambda^y}{y!} ; y \in \{0, 1, 2, 3, \dots\}$$



The Poisson Distribution

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Poisson Distribution: Log-Likelihood Function

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Note: $\sum_{i=1}^n \log(y_i!)$ does not depend on λ

$$\begin{aligned}\frac{\partial \log(L(\lambda \mid \mathbf{y}))}{\partial \lambda} &= -n + \frac{\sum_{i=1}^n y_i}{\lambda} \\ \frac{\partial^2 \log(L(\lambda \mid \mathbf{y}))}{\partial \lambda^2} &= -\frac{\sum_{i=1}^n y_i}{\lambda^2}\end{aligned}$$

But what if we want to model the probability:

How to model *probability* of a binary outcome, $Y \in \{0, 1\}$.

Recall that Y can be:

- Whether a person has a disease or not
- Whether a person will default on a loan or not
- Whether a person will get to college or not

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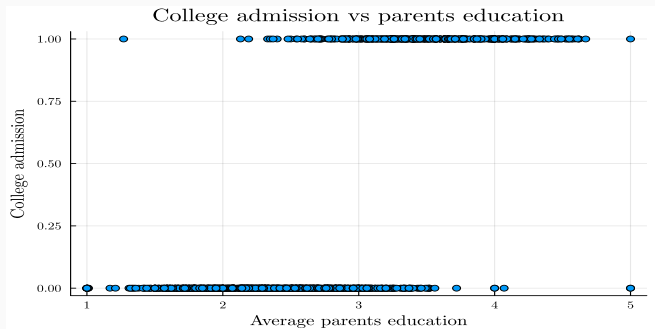
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Making it more concrete:

Suppose you were asked to model the relationship between the probability of getting to college and the (average) parents education.

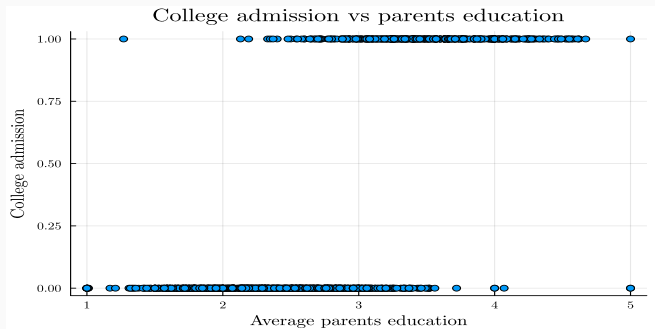
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In this case:

- Y is whether a person gets to college or not
- x is the average parents education

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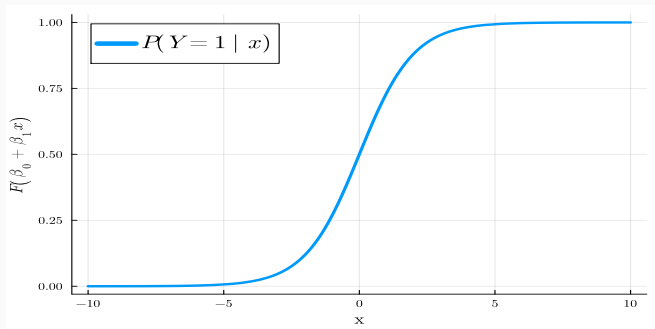
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The Logit Model:

$$\begin{aligned}P(y = 1 \mid x) &= F(\beta_0 + \beta_1 x) \\&= \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \\&= \frac{1}{\frac{1}{e^{\beta_0 + \beta_1 x}} + 1}\end{aligned}$$

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$$\begin{aligned}P(y = 0 \mid x) &= 1 - F(\beta_0 + \beta_1 x) \\&= 1 - \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \\&= \frac{1}{1 + e^{\beta_0 + \beta_1 x}}\end{aligned}$$

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In logit model:

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The likelihood function is:

$$L(\beta_0, \beta_1 | \mathbf{y}, \mathbf{x}) = \prod_{i=1}^N \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{1-y_i}$$

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For example, suppose we have 3 observations:

- $y = [1, 0, 1]$
- $x = [5, 2, 1]$

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$$L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) = \left(\frac{e^{\beta_0 + 5\beta_1}}{1 + e^{\beta_0 + 5\beta_1}} \right) \cdot 1 \cdot 1 \left(\frac{1}{1 + e^{\beta_0 + 2\beta_1}} \right) \cdot 1 \cdot \left(\frac{e^{\beta_0 + \beta_1}}{1 + e^{\beta_0 + \beta_1}} \right)$$

The log likelihood function:

Given:

$$L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) = \prod_{i=1}^N \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{y_i} \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right)^{1-y_i}$$

The log likelihood function is:

$$\log(L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x})) = \sum_{i=1}^N \left\{ y_i \log \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \right\}$$

The log likelihood function

Note that if needed one can split the log likelihood function into two parts:

$$\log(L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x})) = \underbrace{\sum_{i=1}^N y_i \log \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right)}_{\text{The first part}} + \underbrace{\sum_{i=1}^N (1 - y_i) \log \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right)}_{\text{The second part}}$$

The distribution of $(\hat{\beta}_0, \hat{\beta}_1)$

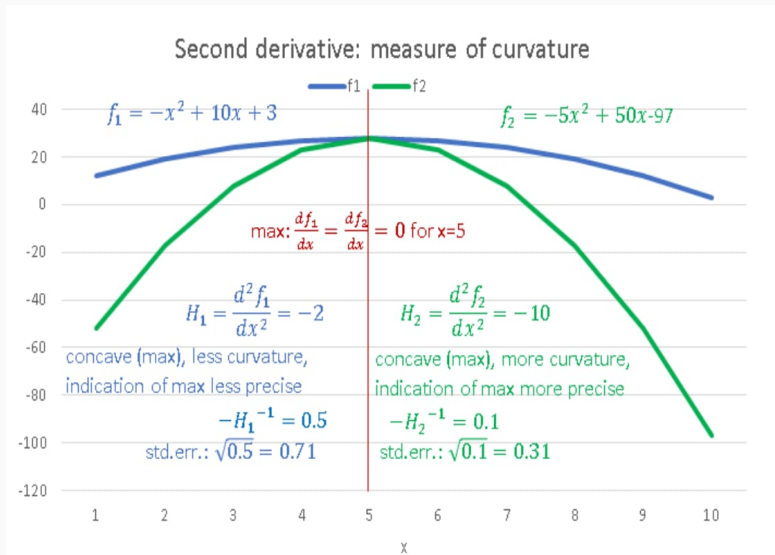
Let $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1)'$ be the MLE from the logit model.

$$\hat{\beta} \sim \mathcal{N}(0, \mathcal{I}(\hat{\beta})^{-1})$$

where $\mathcal{I}(\hat{\beta}) = -H(\hat{\beta})$

In words: the variance of the MLE is given by the negative inverse of the Hessian

Intuition: why is Hessian related to variance?



Logit with more than one independent variable

First note that those forms are equivalent ([click here](#) to see the derivation)

$$\begin{aligned}\log L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) &= \sum_{i=1}^N \left\{ y_i \log \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \right\} \\ &= \sum_{i=1}^N \left[y_i (\beta_0 + \beta_1 x_i) - \log (1 + e^{\beta_0 + \beta_1 x_i}) \right]\end{aligned}$$

Generalization to more independent variables:

Let each observation have a $k \times 1$ vector of regressors $\mathbf{x}_i$ and $\boldsymbol{\beta} \in \mathbb{R}^k$

Then the per-observation log-likelihood becomes:

$$\ell_i(\boldsymbol{\beta}) = y_i(\mathbf{x}_i' \boldsymbol{\beta}) - \log(1 + e^{\mathbf{x}_i' \boldsymbol{\beta}}).$$

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$$\ell_i(\beta) = y_i(\mathbf{x}_i' \beta) - \log(1 + e^{\mathbf{x}_i' \beta}).$$

Summing over all observations:

$$\log L(\beta \mid \mathbf{y}, \mathbf{X}) = \sum_{i=1}^N \ell_i(\beta) = \sum_{i=1}^N \left[y_i \mathbf{x}_i' \beta - \log(1 + e^{\mathbf{x}_i' \beta}) \right].$$

Switching to matrix notation

Stack the regressors into a single matrix:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}'_1 \\ \mathbf{x}'_2 \\ \vdots \\ \mathbf{x}'_N \end{bmatrix}, \quad \mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \mathbf{x}'_1\boldsymbol{\beta} \\ \mathbf{x}'_2\boldsymbol{\beta} \\ \vdots \\ \mathbf{x}'_N\boldsymbol{\beta} \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

Rewrite the summation using matrix products:

$$\sum_{i=1}^N y_i(\mathbf{x}'_i\boldsymbol{\beta}) = \mathbf{y}'(\mathbf{X}\boldsymbol{\beta}),$$

$$\sum_{i=1}^N \log(1 + e^{\mathbf{x}'_i\boldsymbol{\beta}}) = \mathbf{1}' \log(1 + e^{\mathbf{X}\boldsymbol{\beta}}),$$

where $\log(\cdot)$ and $e^{(\cdot)}$ act elementwise.

$$\log L(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}) = \mathbf{y}'(\mathbf{X}\boldsymbol{\beta}) - \mathbf{1}' \log(1 + e^{\mathbf{X}\boldsymbol{\beta}})$$

Gradient and Hessian of the logit log-likelihood

$$\log L(\beta \mid \mathbf{y}, \mathbf{X}) = \mathbf{y}'(\mathbf{X}\beta) - \mathbf{1}' \log(1 + e^{\mathbf{X}\beta}).$$

Define vector of predicted probabilities:

$$\mathbf{p} = \begin{bmatrix} p_1 \\ \vdots \\ p_N \end{bmatrix} = \begin{bmatrix} \frac{e^{x_1' \beta}}{1 + e^{x_1' \beta}} \\ \vdots \\ \frac{e^{x_N' \beta}}{1 + e^{x_N' \beta}} \end{bmatrix}.$$

Gradient(see [derivation](#)):

$$\nabla_{\beta} \log L(\beta \mid \mathbf{y}, \mathbf{X}) = \mathbf{X}'(\mathbf{y} - \mathbf{p})$$

Hessian:

$$H(\beta) = \nabla_{\beta}^2 \log L(\beta \mid \mathbf{y}, \mathbf{X}) = -\mathbf{X}'\mathbf{W}\mathbf{X}$$

$$W = \text{diag}(p_i(1 - p_i)), \quad p_i = \frac{e^{x_i' \beta}}{1 + e^{x_i' \beta}}.$$

The log likelihood function (compact form)

Recall:

$$\log L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) = \sum_{i=1}^N \left\{ y_i \log \left(\frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \right) \right\}.$$

Define $\eta_i = \beta_0 + \beta_1 x_i$. Then:

$$\begin{aligned} \log L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) &= \sum_{i=1}^N \left\{ y_i \log \left(\frac{e^{\eta_i}}{1 + e^{\eta_i}} \right) + (1 - y_i) \log \left(\frac{1}{1 + e^{\eta_i}} \right) \right\} \\ &= \sum_{i=1}^N \{ y_i [\log(e^{\eta_i}) - \log(1 + e^{\eta_i})] + (1 - y_i) [-\log(1 + e^{\eta_i})] \} \\ &= \sum_{i=1}^N \left\{ y_i \eta_i - \underbrace{[y_i + (1 - y_i)]}_{=1} \log(1 + e^{\eta_i}) \right\} \end{aligned}$$

Which gives the **compact log-likelihood with only one exponential**:

$$\log L(\beta_0, \beta_1 \mid \mathbf{y}, \mathbf{x}) = \sum_{i=1}^N \left[y_i (\beta_0 + \beta_1 x_i) - \log(1 + e^{\beta_0 + \beta_1 x_i}) \right]$$

Gradient of $\log L(\beta \mid \mathbf{y}, \mathbf{X})$

Start from the summation form & differentiate w.r.t. β :

$$\log L(\beta \mid \mathbf{y}, \mathbf{X}) = \sum_{i=1}^N \left[y_i \mathbf{x}'_i \beta - \log(1 + e^{\mathbf{x}'_i \beta}) \right].$$

First term:

$$\frac{\partial}{\partial \beta} (y_i \mathbf{x}'_i \beta) = y_i \frac{\partial}{\partial \beta} (\mathbf{x}'_i \beta) = y_i \mathbf{x}_i.$$

Second term:

$$\frac{\partial}{\partial \beta} \log(1 + e^{\mathbf{x}'_i \beta}) = \frac{1}{1 + e^{\mathbf{x}'_i \beta}} \cdot e^{\mathbf{x}'_i \beta} \cdot \frac{\partial}{\partial \beta} (\mathbf{x}'_i \beta) = \frac{e^{\mathbf{x}'_i \beta}}{1 + e^{\mathbf{x}'_i \beta}} \mathbf{x}_i = p_i \mathbf{x}_i$$

Putting it together:

$$\nabla_{\beta} \log L(\beta \mid \mathbf{y}, \mathbf{X}) = \sum_{i=1}^N [y_i \mathbf{x}_i - p_i(\beta) \mathbf{x}_i] = \sum_{i=1}^N \mathbf{x}_i (y_i - p_i(\beta)).$$

In matrix notation, with $\mathbf{p} = (p_1(\beta), \dots, p_N(\beta))'$:

$$\nabla_{\beta} \log L(\beta \mid \mathbf{y}, \mathbf{X}) = \mathbf{X}'(\mathbf{y} - \mathbf{p}).$$